

Soal nomor 1

```
#include <iostream>
#include <iomanip>
using namespace std;

struct Mahasiswa {
    string nama;
    string nim;
    float uts;
    float uas;
    float tugas;
    float nilaiAkhir;
};

float hitungNilaiAkhir(float uts, float uas, float tugas) {
    return 0.3 * uts + 0.4 * uas + 0.3 * tugas;
}

int main() {
    Mahasiswa mhs[10];
    int n;

    cout << "Jumlah mahasiswa (maks 10): ";
    cin >> n;

    for(int i = 0; i < n; i++) {
        cout << "\nData Mahasiswa ke-" << i+1 << endl;

        cin.ignore();
        cout << "Nama : ";
        getline(cin, mhs[i].nama);

        cout << "NIM : ";
        getline(cin, mhs[i].nim);

        cout << "UTS : ";
        cin >> mhs[i].uts;

        cout << "UAS : ";
        cin >> mhs[i].uas;

        cout << "Tugas : ";
        cin >> mhs[i].tugas;

        mhs[i].nilaiAkhir = hitungNilaiAkhir(
            mhs[i].uts,
            mhs[i].uas,
            mhs[i].tugas
        );
    }
}
```

```

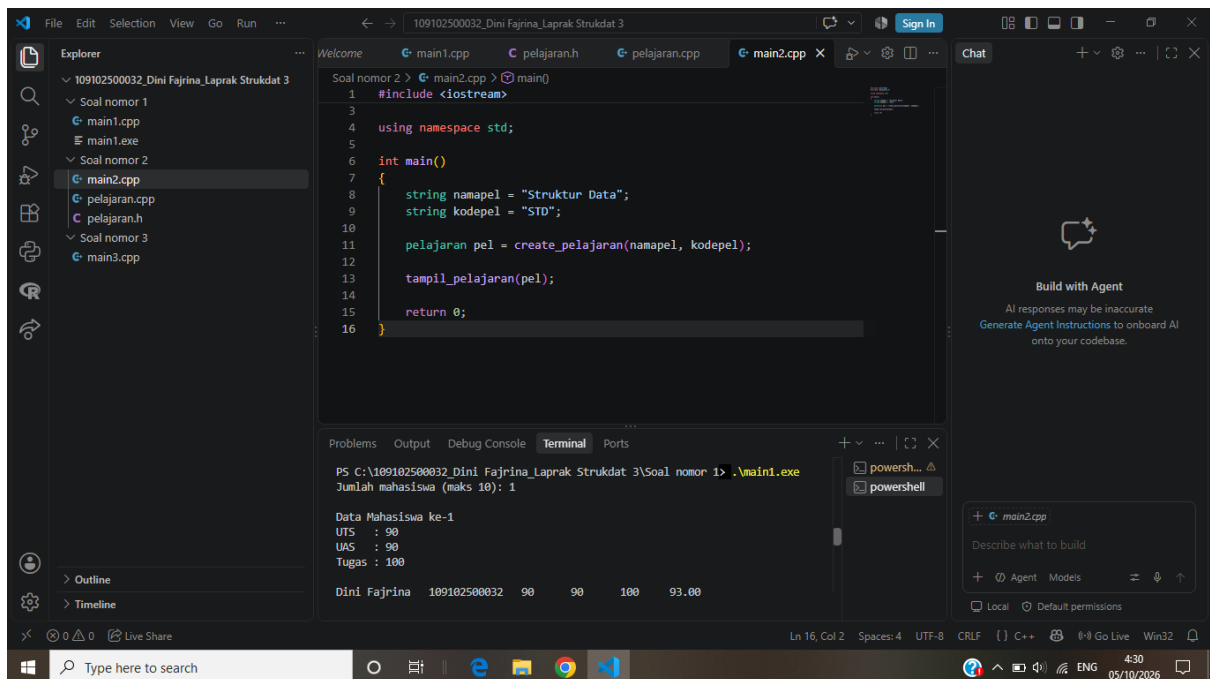
}

cout << "\n===== DATA MAHASISWA =====" << endl;
cout << left
    << setw(15) << "Nama"
    << setw(15) << "NIM"
    << setw(8) << "UTS"
    << setw(8) << "UAS"
    << setw(8) << "Tugas"
    << setw(12) << "Nilai Akhir"
    << endl;

for(int i = 0; i < n; i++) {
    cout << left
        << setw(15) << mhs[i].nama
        << setw(15) << mhs[i].nim
        << setw(8) << mhs[i].uts
        << setw(8) << mhs[i].uas
        << setw(8) << mhs[i].tugas
        << setw(12) << fixed << setprecision(2) << mhs[i].nilaiAkhir
        << endl;
    }

return 0;
}

```



Soal nomor 2

- Pelajaran.h

```
#ifndef PELAJARAN_H
#define PELAJARAN_H

#include <string>
using namespace std;

struct pelajaran {
    string namaMapel;
    string kodeMapel;
};

pelajaran create_pelajaran(string namapel, string kodepel);
void tampil_pelajaran(pelajaran pel);

#endif
```

- Pelajaran.cpp

```
#include <iostream>
#include "pelajaran.h"

using namespace std;

pelajaran create_pelajaran(string namapel, string kodepel)
{
    pelajaran pel;

    pel.namaMapel = namapel;
    pel.kodeMapel = kodepel;

    return pel;
}

void tampil_pelajaran(pelajaran pel)
{
    cout << "nama pelajaran : " << pel.namaMapel << endl;
    cout << "nilai : " << pel.kodeMapel << endl;
}
```

- main.cpp

```
#include <iostream>
#include "pelajaran.h"
```

```

using namespace std;

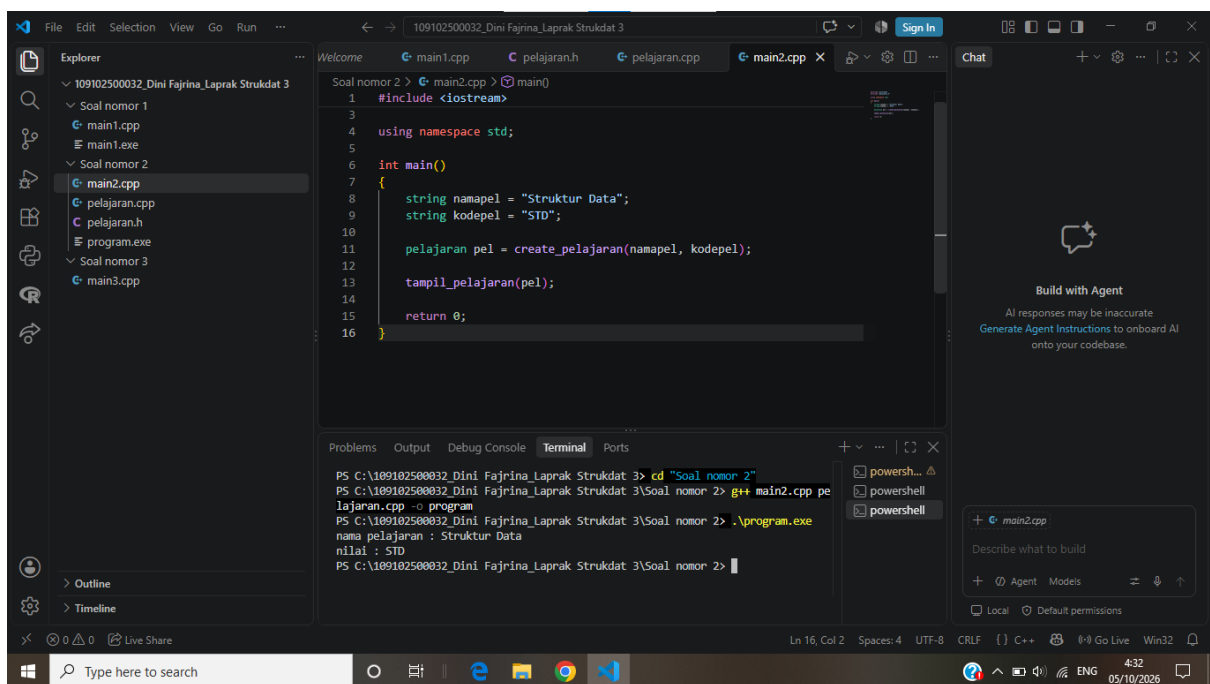
int main()
{
    string namapel = "Struktur Data";
    string kodepel = "STD";

    pelajaran pel = create_pelajaran(namapel, kodepel);

    tampil_pelajaran(pel);

    return 0;
}

```



Soal nomor 3

```

#include <iostream>
using namespace std;

void tampilArray(int arr[3][3]) {
    for(int i = 0; i < 3; i++) {
        for(int j = 0; j < 3; j++) {
            cout << arr[i][j] << " ";
        }
        cout << endl;
    }
}

void tukarArray(int A[3][3], int B[3][3], int baris, int kolom) {
    int temp = A[baris][kolom];
    A[baris][kolom] = B[baris][kolom];
}

```

```

    B[baris][kolom] = temp;
}

void tukarPointer(int *p1, int *p2) {
    int temp = *p1;
    *p1 = *p2;
    *p2 = temp;
}

int main() {
    int A[3][3] = {
        {1,2,3},
        {4,5,6},
        {7,8,9}
    };

    int B[3][3] = {
        {9,8,7},
        {6,5,4},
        {3,2,1}
    };

    cout << "Array A:" << endl;
    tampilArray(A);

    cout << "\nArray B:" << endl;
    tampilArray(B);

    tukarArray(A, B, 1, 2);

    cout << "\nSetelah pertukaran posisi [1][2]" << endl;

    cout << "Array A:" << endl;
    tampilArray(A);

    cout << "\nArray B:" << endl;
    tampilArray(B);

    int x = 10, y = 20;
    int *p1 = &x;
    int *p2 = &y;

    cout << "\nSebelum tukar pointer:" << endl;
    cout << "x = " << x << ", y = " << y << endl;

    tukarPointer(p1, p2);

    cout << "Sesudah tukar pointer:" << endl;
    cout << "x = " << x << ", y = " << y << endl;

```

```
return 0;
}
```

